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Introduction to health measurement scales

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Abstract

Both research and clinical decision making rely on measurement scales. These scales vary with regard to their psychometric properties, ease of administration, dimensions covered by the scale, and other properties. This article reviews the main psychometric characteristics of scales and assesses their utility.
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Introduction

There is no single variable that can be used to describe health, and health cannot be measured directly. Health measurement requires several steps and involves the evaluation of several health-related indicators.

Rating scales are used in numerous settings to measure various aspects of health such as different symptoms or the presence of a particular trait. Health measurement scales can be classified in (at least) three ways, according to their function, description, and methodology. Functional classification focuses on the application of methods and how they are used, such as Bombardier and Tugwell's [1] classification of diagnostic, prognostic, and evaluative health measurements; however, others [2] have argued that this classification ignores the way scales are actually used in practice. Descriptive classification of health measurements is concerned with the range of topics covered by a particular measurement. For example, one might focus on a particular organ system, a diagnosis, or a broader concept such as anxiety or quality of life. Another distinction can be between broad classification of generic health measures and specific instruments. Specific instrument can be concerned with not only a particular disease, but also a particular target population, such as children. Methodological classification distinguishes among rating scales, questionnaires, indices, and subjective vs. objective measures.

Whether rating scales are to be used in a research project or to make clinical decisions, it is essential to evaluate how well they perform. By how well, we mean how much random error is present in the measurement (i.e., its reliability) and whether the scores give us meaningful information about the respondent (the validity of the instrument). A third measure of performance addresses the issue of whether it is feasible to use the instrument for a particular purpose. In this article, we will give an introduction to some of the properties of rating scales, the concept of validity and reliability. Those who are interested in the details of constructing measurement scales are referred to more comprehensive texts [2–4].

Scale development can be approached in two ways: questions may be chosen from an empirical or a theoretical viewpoint [5]. With the empirical approach, a large number of questions are tested and statistical procedures are used to select the ones that best predict the outcome of interest. However, the disadvantage of this method is that it is difficult to interpret why individuals answering a certain question in a certain way tend to have different outcomes. Questions in the Health Opinion Survey [6] were selected because they distinguished between those who do and do

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not have psychiatric problems. However, debates over what exactly the scale measures are still continue. Scales developed entirely from an empirical stance may have clinical value, but they do not advance our understanding of the underlying phenomena. The alternative strategy is to select questions that are thought to be relevant from a standpoint of a particular theory, such as the McGill Pain Questionnaire [7]. In psychology, at least, the trend over the past 50 years has been a move toward theoretically derived instruments [2].

Items on a scale

Items on a scale can come from several different sources: existing scales, reports of individuals' subjective experiences, clinical observations, expert opinion, research findings, and theory. One should be aware of the strengths and weaknesses of each source when considering a scale for a particular use. The advantage of using existing items from older scales is that items have probably already gone through a rigorous process of assessment and are, therefore, more likely to be useful. It may thus save time and work rather than construct new items. However, outdated terminology may render some older items useless.

Patients experiencing a trait or disorder can be excellent sources of scale items, especially when the interest lies in the more subjective elements of the trait. *Focus groups* and *key informant interviews* are techniques that can be used to acquire patients' viewpoints in a systematic manner [8].

Clinical observation as a means of developing scale items can be useful, as these observations precede any theory, research, or expert opinion. Scales developed in this manner can be seen as a structured way of assembling clinical observations. The major disadvantage, however, is that the clinician developing the scale might have been wrong in his/her observation. If a scale is based on unreplicated findings, it leads to a useless scale. For example, a scale that is based on the erroneous observation that the incidence of epilepsy is lower in the schizophrenic population is destined to failure. Moreover, the clinician observes a particular phenomenon on a limited sample of patients and, therefore, may miss other relevant factors that would be apparent in another population. One way to overcome the problem of mistaken observations is to use the judgment of not just one but a panel of experts. The advantage of this approach is that an expert panel probably represents the most recent views in a topic. A note of caution is in order, however, as there are no rules on how and how many experts have to be chosen and how opinions have to be synthesized. If the selected experts all share the opinion and perhaps biases regarding the domain to be measured, then using a panel of experts does not provide additional advantage to using the views of just one person.

Research findings from literature reviews of previous studies in the area or new research carried out for developing a scale can be another source of items. An example is a subset of a scale developed to differentiate between irritable bowel syndrome and organic bowel disease [9]. It consists of laboratory values and clinical history that were chosen on the basis of previous research, indicating difference between irritable bowel syndrome and organic disease regarding these variables.

As mentioned previously, a set of clinical and/or laboratory observations forming a theory about differences in patients might also provide items. In this context, we should not only think of formal theories, but also more vague ideas such as the notion that patients believing in the efficacy of a treatment will be more compliant. The weakness of using "theories" in item selection is the possibility of using a wrong model, and this may only be apparent later when the validity of the scale is assessed.

Criteria to identify useful items

Not every item intended for a scale will perform well; therefore, several aspects of items have to be checked to decide which are likely to be useful.

It is important to use clear, comprehensible language. Very often, technical or jargon terms are used (e.g., *stool*, *shock*, or *cardiovascular*), which would be fine if the scale is to be used on health professionals but not if lay people are the intended respondents. Since people are different in their reading ability, items should not require more than very basic reading skills. The scales should be tested on the target group to verify that the used terms are understandable.

Another potential problem related to language that can result in unintended responses is ambiguity, which can be caused, for example, by using vague terms. The answer to the question, "Have you been in hospital recently?" depends on how the respondents interpret "recently" and if they differentiate among being an inpatient, outpatient, or even a visitor. We should also check for and avoid items that incorporate two questions (e.g., "I feel sad and lonely"), because some people would answer "yes" if one part of the question applies to them, while others will say "yes" only if both apply.

Terms that might offend or prejudice people should be avoided. Items such as "Do physicians make too much money" may indicate to the respondent what the desirable answer would be.

Items that are very likely (more than 90% of the time) answered in one way or the other are not very useful. If everyone answers "yes" to a question, then that item does not contribute to discriminating between individuals who have a certain characteristic and those who do not. Furthermore, such questions can introduce unnecessary measurement error caused by random responding or other reasons for not giving the true answer.

Reliability

Before one can start using an instrument, it should be established that it is measuring "something" in a

reproducible manner; that is, if the measurement is repeated by different observers, or on different occasions, or by a similar (parallel) test, then the results should be comparable [2]. Assuming that the person has not changed, we would expect to arrive at similar scores at two different times. There are different indices to measure reliability, and not all are applicable to a given scale. It is not necessary, for example, to show interrater reliability if a test is self-administered.

It is also important to emphasize that reliability (and, as we will discuss, validity) is not a fixed property of a scale. Rather, it is a function of the instrument, the group with which it is being used, and the circumstances. It is wrong to talk about *the* reliability of a scale, as opposed to the reliability of a scale used with a specific population for a given purpose. A scale that is reliable in one set of circumstances may not be reliable under different conditions.

Internal consistency

This is a measure based on a single administration of a test and therefore is easy to obtain. It measures the average correlation among all items in the measure. One expects that scores on items tapping the same underlying dimension would correlate well. A low internal consistency could mean that the items measure different attributes or the subjects' answers are inconsistent. Internal consistency can be measured by Cronbach's α [10], which is derived from the Kuder–Richardson formula 20 [11], or the split-half method [2]. These measures do not take into account variations in time or from observer to observer and therefore yield an optimistic estimate of the true reliability of the test. A major problem with these indices is that they are sensitive not only to the internal consistency of the scale, but also to its length. α will be high if there are more than about 15 items, irrespective of the correlation among them [12], and for longer scales, other indices should be used, such as the mean interitem correlation [13].

Test–retest reliability

If a patient's status in respect to what is being measured does not change between two time points, then measurements taken at these times should be the same, or very similar. The time interval between measurements and the stability of the construct over time will influence our interpretation of a test–retest reliability estimate. That is, we expect traits such as introversion to be more stable over time than more transient states, such as depression. In order to avoid changes over time, the two measurements should be temporally closed to each other. However, this may result in recalling the previous answers rather than giving an independent response, especially with scales consisting of few items. The usual test–retest interval is between 10 and 14 days. In general, test–retest reliability coefficients above 0.9 are considered high and the minimum for clinical scales, and between 0.7 and 0.8 are acceptable for research tools.

Interrater reliability

When a scale is completed by a rater, and not by the patient him or herself, then different raters assessing the same individual should obtain similar scores. Interrater reliability is measured with a coefficient between 0 and 1 and, in general, follows the criteria for test–retest reliability. When the raters evaluate the person at different times, an additional source of variation is introduced, namely, the change in the patient between the two ratings. Hence, in these cases, a lower interrater reliability coefficient may be acceptable.

Although test–retest and interrater reliability are usually measured with Pearson's r, it is better to use the intraclass correlation based on a two-way repeated measures analysis of variance looking at absolute agreement, since this is sensitive to any bias between or among the raters or times.

Validity

Validity is concerned with the meaning and interpretation of the scores. In other words, validity guides us as to what conclusions can be made about people with a given score. If, for example, we draw on a scale to measure degree of low-back pain, then we would like to be sure that people who score higher actually have more low-back pain. Whether this is the case is a question of validity. It may be that the scale measures something else, such as the degree of pain from other sources or tendency to complain.

Assessment of validity can be done in various ways, each evaluating different aspects of a scale, and it should be thought of as an ongoing process rather than a definite conclusion. As with reliability, validity is not an inherent property of the measure but an interaction of the scale, the group being tested, and the conditions. Although the conceptualization of validity testing has shifted from the "types" of validity to seeing all types as subsets of *construct validity*, it is useful for didactical and historical reasons to explain validity in the context of its traditional categorization into the so-called *three Cs* of validity: content, criterion, and construct validity [14]. However, to be consistent with the newer definition of validity, we will refer, for example, to criterion *validation* rather than criterion *validity*, reflecting that it is a method of determining validity rather than a type of validity.

Criterion validation

Criterion validation consists of correlating the new scale with a widely accepted measure of the same characteristics —the "gold standard." If the comparison of the two scales is done at the same time, it is called *concurrent validation*. This situation arises usually when some aspect of the new scale (e.g., its cost or invasiveness) is thought to be superior to the gold standard. In this case, we would expect the

correlation to be high (≥0.8). On the other hand, if the new scale is believed to be better because it is more valid than the old one, then we do not want to see a very high correlation, since that would mean the new scale is not any different. If the correlation is too low (<0.30), then the measures are not related to each other, indicating that they measure different characteristics.

When a scale is compared to a criterion that is measured later, the new test is evaluated by how well it predicts the criterion score. This type of validity testing is called *predictive validation* and is used often with diagnostic tests and school admission tests, where one is interested before students are admitted to school how well they will perform, that is, whether or not they graduate a number of years later.

Content validation

A scale measuring a patient's level of depression must cover all aspects of depression and should not include items that are unrelated to that construct. Unlike with other forms of validation, there is no correlation coefficient or some other statistic that can be used to measure content validation. The test developer has to evaluate whether all relevant aspects of a trait or disorder are included in a scale and if there are irrelevant ones. It is worth noting that there is usually an inverse relationship between content validation and internal consistency. A test intended to measure a heterogeneous trait or disorder might have a relatively low internal consistency, which can be increased by eliminating items that show low correlation with other items. However, by eliminating items, fewer aspects of the trait are addressed by the scale, leading to reduced content validation.

Construct validation

Unless we measure readily observable physical variables, we rely on the measurement of certain surrogate attributes that we think fit into our theory of a certain concept. In psychology, these abstract variables are called hypothetical constructs [15]. For example, we could measure heart rate, sweatiness, and difficulty in concentrating because our "theory" tells us that these are observable manifestations of the underlying process of anxiety, our construct. Construct validation is usually heavily relied upon in situations when there is no criterion with which a scale can be compared.

In order to establish construct validity, one has to generate predictions based upon the hypothetical construct, and these predictions can be then tested to give support to the validity of the scale. As noted above, though, because saying that the new test should correlate with another test of the same construct (what had been called *criterion validation*) is itself a prediction, all validational studies are now subsumed under *construct validation*.

An instrument's ability to measure change, called *sensitivity to change*, is another useful component of scale evaluation. It has been suggested to be a distinct attribute of a scale, but many researchers consider it conceptually a part of validity [2,16,17].

Face validation

One more type of validity can be added to the "three C's," which deals with whether the scale items appear on the surface to be measuring what the scale intends to measure. *Face validation* is a concept similar to content validation, and it has to be evaluated similarly by a subjective judgment of experts. Face validity is desirable most of the time, since questions that are perceived irrelevant may cause respondents to not take them seriously or refuse to answer. However, in certain situations, the opposite might be favored. If respondents are likely to falsify their answers, then it may be useful to avoid scales with high face validity.

Utility

A measure that is reliable and valid can still be impractical for use. It may, for example, take a long time to complete, may require excessive resources to score, or may require training interviewers who would administer the scale. It is usually the case that longer tests tend to be more reliable and valid than shorter ones, but for the sake of improved utility, decreasing the time needed to complete a test might be advantageous.

Over the past few decades, a new model of test construction has been widely adopted in the area of education and for "high-stakes" tests, such as licensing examinations. Called item response theory (IRT), it purportedly overcomes some of the limitations of scales developed with the so-called classical test theory (CTT), such as the dependence of the psychometric properties of a scale on the normative sample, the fact that the score rarely meets the assumptions of being on an interval scale, and so on [18]. However, IRT is rarely used in developing scales used to assess various aspects of health and illness, because scales developed using CTT are adequate for their purpose [2], and it has been argued that the added work required to use IRT is rarely worth the effort.

Summary

Scales and questionnaires are an integral part of clinical practice and research. However, they are not all created equally. To be useful, instruments must demonstrate good psychometric properties, such as reliability and validity, and be in a format that patients find easy to use.

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Suggested further reading:

Books and articles about scale development and psychometric theory:

DeVillis RF. Scale development: theory and applications. 2nd ed. 2003. Sage: Thousand Oaks, CA.

An excellent text, albeit oriented more toward psychologists.

Streiner DL. A checklist for evaluating the usefulness of rating scales. Can J Psychiatry 1993;38:140–148.

This article summarizes the properties to look for in evaluating scales.

Streiner DL, Norman GR. Health measurement scales: a practical guide to their development and use. 4th ed. 2008. Oxford: Oxford University Press.

This book is oriented more toward health-related attributes.

Good compendia of tests used in health:

Fischer J, Corcoran KJ. Measures for clinical practice: a sourcebook. 4th ed. 2007. New York: Oxford University Press.

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These books provide the actual scales used in a variety of health-related conditions.

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